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A newly developed long range positioning table system with a sub-nanometer resolution

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ABSTRACT

Demands for nano-positioning over a long range have recently increased in a variety of industries. This paper presents a long range positioning table system with a sub-nanometer resolution. The table system developed is characterized by a motion error-minimized structure, and the table motion is implemented by a cooperative control of the primary and the secondary tables. Successful implementation of a laser interferometer makes it possible to achieve the Abbe's error free measurement and a long range positioning with a sub-nanometer resolution. Experimental results confirm that the table system achieves sub-nanometer positioning over a 150 mm range.

1. Introduction

Demands for nano-machining and nano-metrology over a relatively large range have recently increased in a variety of industries and sciences such as aerospace, electronics, energy, and astronomy [1–3]. In particular, such demands result in the development of innovative table systems which achieve long range positioning with a sub-nanometer resolution [4]. Although many studies on ultra-precision positioning table system have been performed so far, there were only few studies on table system which could achieve positioning with sub-nanometer resolution in a long travel range [5–10].

This study aims at developing a long range positioning table system with a sub-nanometer resolution. The table system developed is designed based on a new concept and controlled with a new control scheme. In order to achieve a stable and accurate motion control in a long range, an improved homodyne interferometer with sub-nanometer resolution is used as the position feedback sensor. Performance evaluation results confirm that the developed table system in capable of providing a resolution of sub-nanometer in the 150 mm travel range.

2. Proposed design concept of a long range positioning table system with a sub-nanometer resolution

Fig. 1 shows a proposed design concept of a long range positioning table system with a sub-nanometer resolution. Based on the proposed design concept, an innovative table system can be realized, as shown in Fig. 2.

The table system proposed can be characterized by the following functions and structures:

(1) Long range positioning with a sub-nanometer resolution,
(2) A symmetrically designed structure with respect to the motion axis,
(3) Driving at the center of gravity of a moving table,
(4) Hybrid actuator combined a voice coil motor (VCM) with a ball screw,
(5) Motion error-minimized structure,
(6) Noncontact driving and supporting,
(7) Laser interferometer feedback system with frequency doubled signals together with multiple optical paths,
(8) Cooperative control for keeping a constant relative displacement between the primary and the secondary tables,
(9) Thermal deformation-minimized structure.

The laser interferometer feedback system enables sub-nanometer positioning of the primary table while keeping a constant relative displacement between the primary and the secondary tables. The design concept proposed here is successfully applicable to the table systems which achieve sub-nanometer positioning over a stroke of several hundred millimeters.

3. Developed long range positioning table system with a sub-nanometer resolution

3.1. Structural configuration

Fig. 3 shows the structural configuration of the developed table system. Specifications of the table system are given in Table 1. The moving table is just a simple plate with a rectangular hole at the center to install a VCM. The table has porous air bearing pads on the bottom and both side surfaces. Neodymium magnets for preloading in the vertical direction are fixed on the bottom of the primary table. This structure can make eddy current loss minimum because the relative velocity between the primary and secondary tables is almost zero. In addition, air tube from the primary table is fixed on the secondary table to eliminate the disturbances from the tube

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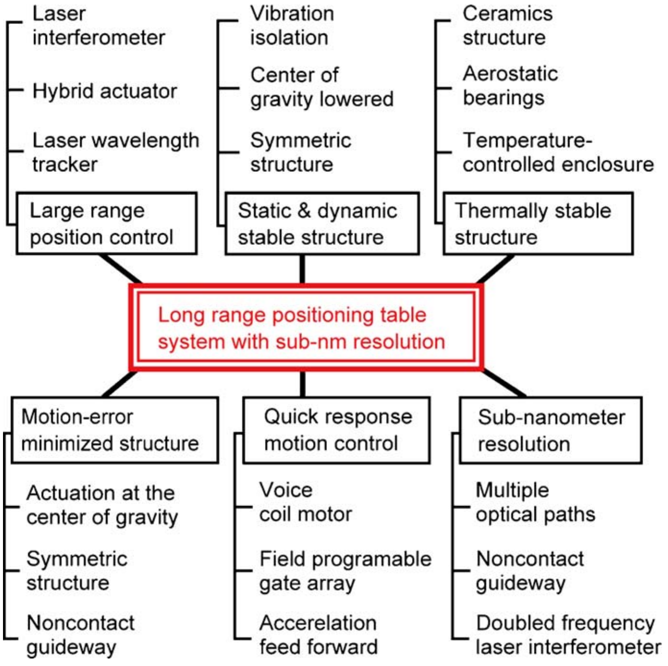

Fig. 1. Design concept of the long range positioning table system with sub-nanometer resolution.

during a long range positioning. The fundamental structural modules, such as a moving table, aerostatic guideways, and a base plate, were made of alumina ceramics (Al_2O_3) to reduce the thermal deformation and improve the stiffness. Sub-nanometer positioning over a long range was achieved by combining the VCM with the ball screw feed drive system.

3.2. Laser interferometry for achieving sub-nanometer motion control

A position feedback sensor for achieving the motion control of a sub-nanometer resolution requires a measuring resolution of less

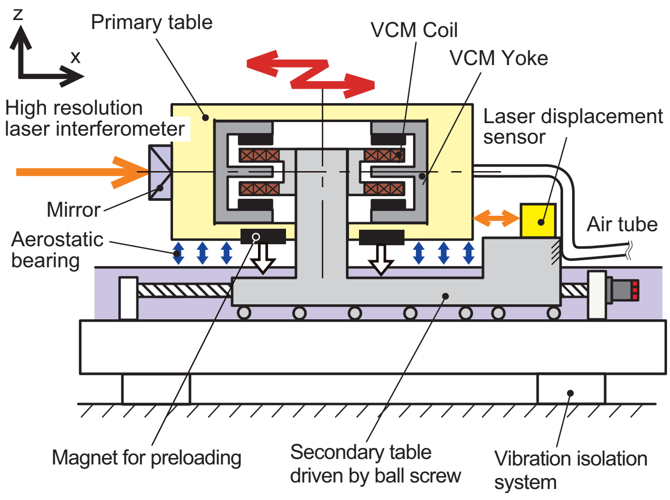

Fig. 2. Structural concept of the table system proposed.

than sub-nanometer. In this study, a homodyne interferometer with sub-nanometer resolution was used for achieving a stable motion control over a long range. In order to improve the resolution, the frequency of output signals obtained from the photo detector was doubled according to Eq. (1) which converted from original signals to frequency doubled signals by using an originally designed analog circuit.

$$\begin{aligned} \sin 2x &= 2\cos x \cdot \sin x \\ \cos 2x &= (\cos x + \sin x)\cdot(\cos x - \sin x) \end{aligned} \qquad (1)$$

In general, the use of multiple optical paths improves the resolution of the interferometer. In this study, the laser paths between both the objective table and the interferometer were repeated four times to improve the sensor resolution, as shown in Fig. 4. In addition, the refractive index of air was compensated by an originally developed wavelength tracker. In consequence, a measuring resolution of 9.7 pm., which was a quarter of the original resolution, could be obtained.

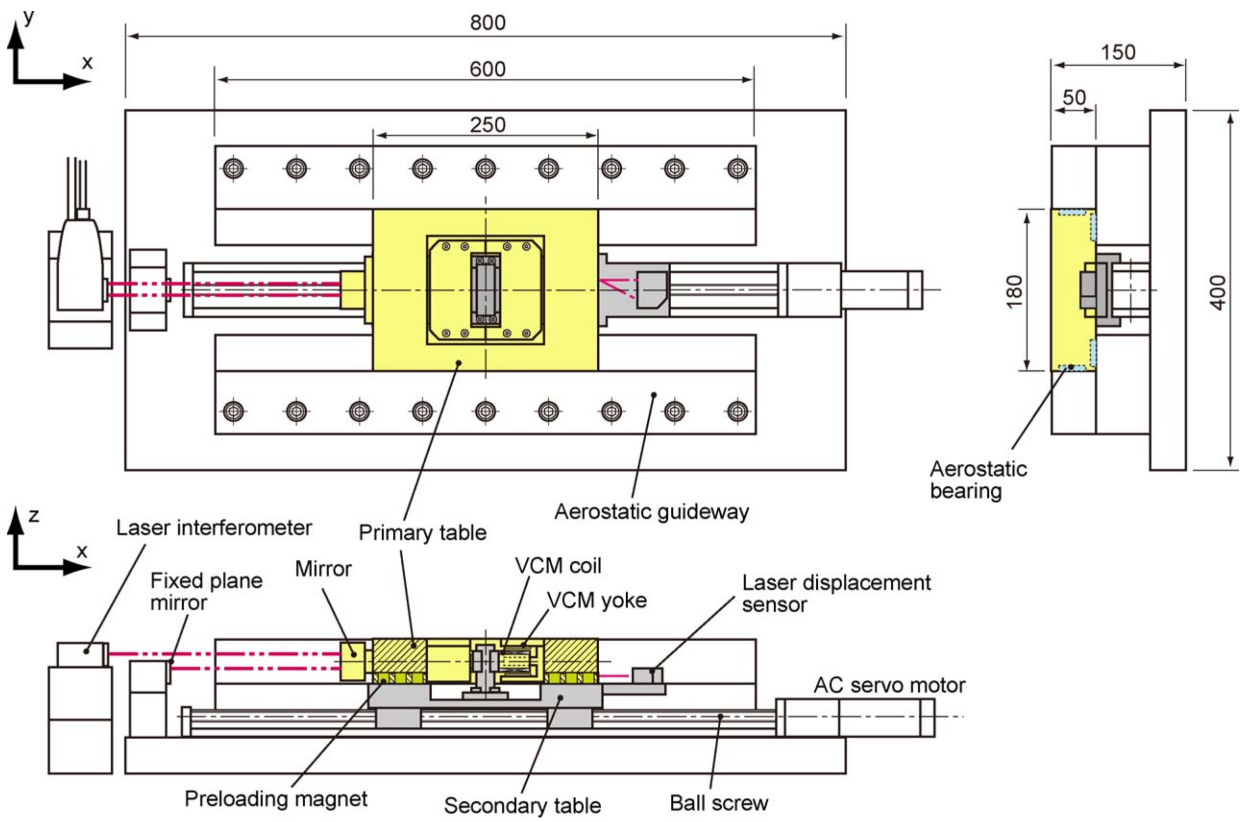

Fig. 3. Structural configuration of the table system developed.

Table 1
Specifications of the table system developed.

Full stroke	300 mm
Stroke of VCM	±2.5 mm
Mass of table	7.8 kg
Max. velocity	220 mm/s
Max. acceleration	5.35 m/s^2
Sensor resolution	9.7 pm
Control frequency	33 kHz
Stiffness	
Vertical	326 N/μm
Horizontal	326 N/μm
Straightness	
Vertical	0.10 μm/100 mm
Horizontal	0.18 μm/100 mm

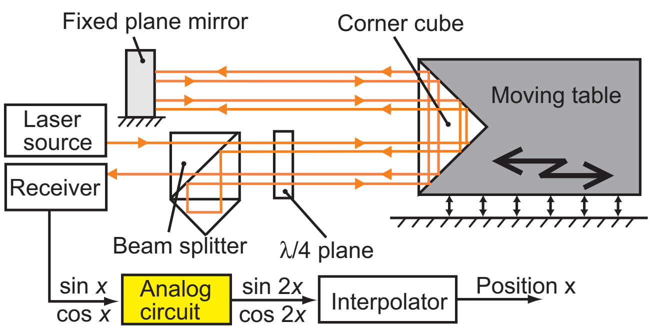

Fig. 4. System configuration of the four paths interferometer developed.

3.3. Sub-nanometer motion control system

Fig. 5 shows a block diagram of the sub-nanometer motion control system, which consists of the primary table position control system and the relative displacement control system. The full closed loop control system with the laser interferometer feedback was implemented with the PID and the acceleration feedforward compensators. These compensators made it possible to realize stable and high speed positioning. The relative displacement control system was implemented to be kept a constant relative displacement during positioning. The acceleration feedforward compensator was implemented in the primary table position control system so as to eliminate the tracking errors. Use of a field programmable gate array (FPGA) with a clock rate of 40 MHz enabled the control loop to improve the response speed.

Fig. 6 shows the table system developed. In particular, the table position measured by the interferometer depends on the external disturbances such as vibration and ambient temperature. Therefore, the overall table structure was mounted on a vibration isolation system and installed in an enclosure which was controlled within a temperature range of 22 ± 0.2 °C.

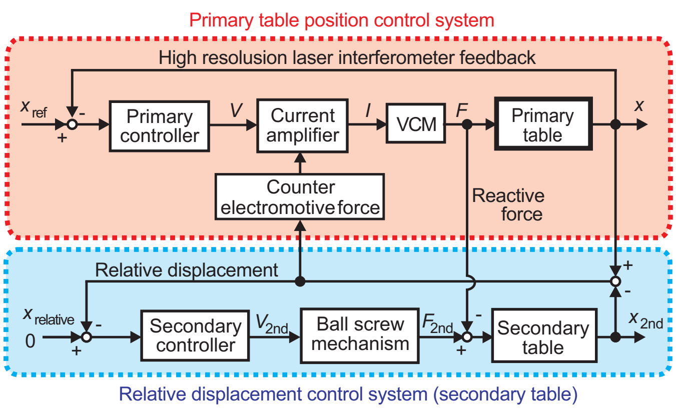

Fig. 5. Block diagram of the motion control system developed.

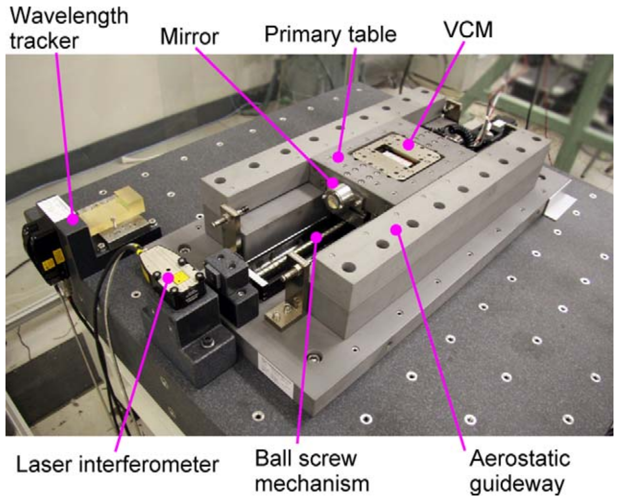

Fig. 6. Appearance of the table system developed.

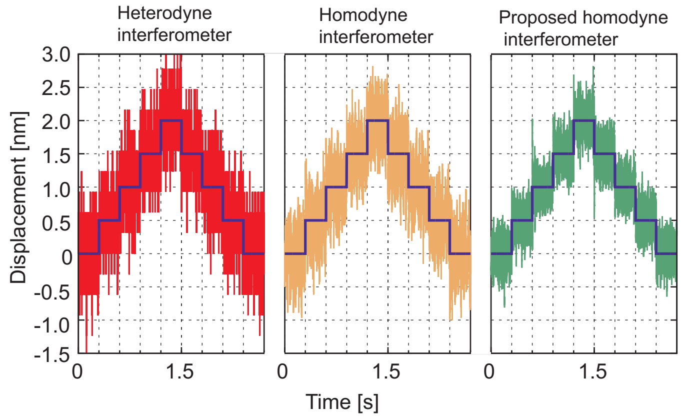

Fig. 7. Typical examples of sub-nm stepwise response.

4. Performance evaluation results of the positioning table system

Sub-nanometer stepwise responses were investigated by using three kinds of laser interferometers; a double path heterodyne interferometer, a commercially available homodyne interferometer and the proposed homodyne interferometer. Fig. 7 shows typical examples of sub-nanometer stepwise responses. The stepwise response results show the effectiveness of the proposed interferometer, as shown in Fig. 8. Although the output signals include ±0.3 nm, both 0.5 nm and 0.3 nm stepwise responses can be clearly observed, as shown in Fig. 8(a) and (b). In addition, the point-to-point (PTP) motion of 100 nm was performed at a feedrate of 100 nm/s and an acceleration of 200 nm/s^2. The table achieved a tracking error of less than ±2 nm, as shown in Fig. 9.

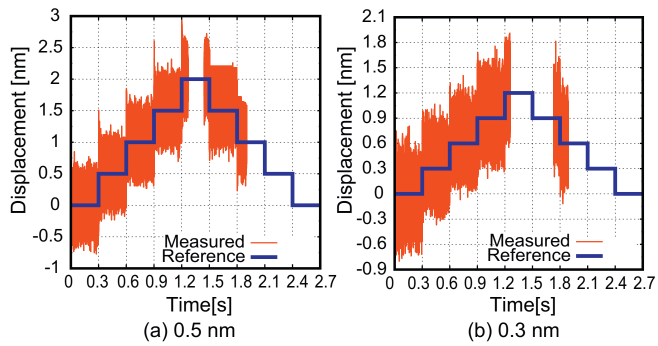

Fig. 8. Sub-nanometer stepwise response.

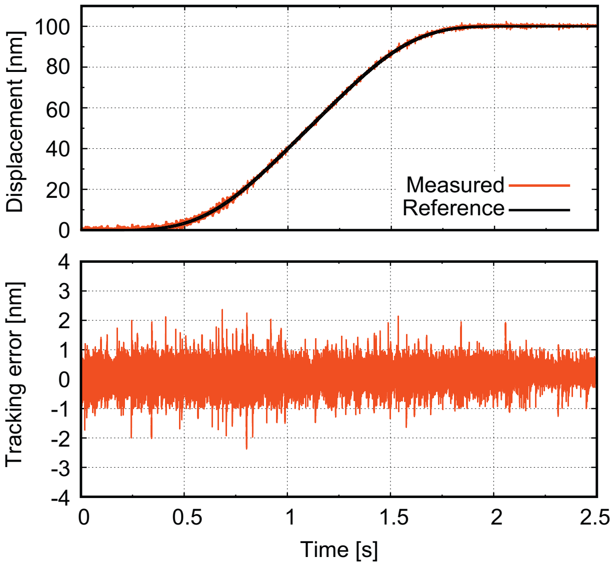

Fig. 9. Tracking error at a feedrate of 100 nm/s.

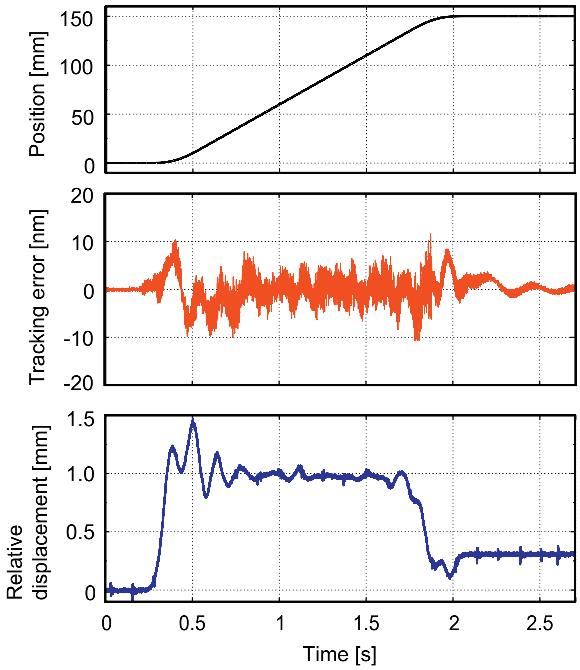

Fig. 10. Tracking error in a long range of 150 mm.

The table system developed provided sub-nanometer positioning capability over a long range larger than the stroke of VCM. The high speed PTP motion of 150 mm was performed with a jerk continuous trajectory. Fig. 10 shows the position command, the tracking error and the relative displacement between the primary and the secondary tables at a feedrate of 100 mm/s and an acceleration of 500 mm/s^2. Although the maximum relative displacement of 1.5 mm caused between both tables, the tracking error was negligibly small. This result confirmed that no tracking error of the secondary table influences positioning accuracy of the primary table. In consequence, the table system developed achieves tracking accuracy of a nanometer scale in a long range.

5. Conclusions

This paper presented a newly developed long range positioning table system with a sub-nanometer resolution. In addition, the performance was evaluated. As a result, the following conclusions could be drawn.

(1) The innovative table system was developed for achieving positioning with a 0.3 nm resolution in a 150 mm range.
(2) The performance evaluation results confirmed that the table system developed provides the positioning and tracking accuracy of a sub-nanometer order.

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